

4. Electronic Controller Introduction

4.1 Electronic controller

RT-----Room Temperature.

IPT-----Indoor Pipe (Coil) Temperature.

ST-----Indoor Setting Temperature.

OPT--- Outdoor Pipe Temperature.

CRT--- Compensated Room Temperature.

IDU--- Indoor unit

ODU--- Outdoor unit.

Note: When AC finishing installation, because of the air ventilation and the distance of temperature test sensor to different location of the house, the temperature for IDU PCB control need compensation.

1) . **Cooling mode.** CRT=RT;

2) . **Heating mode.** CRT=RT-3°C ----- Split AC.

4.1.1 Auto mode

5.2.1.1. The set temperature can be adjusted from 16-31°C (61-88°F) on auto mode, the operation of fan speed and vane position according to preset.

5.2.1.2. Operation

When unit set to auto mode, it will work on cooling, heating or fan mode totally according to Δt --the temperature difference between RT and ST shown as table:

Mode	$\Delta t = RT - ST$
COOLING	$\Delta t > 1^{\circ}\text{C}$
FAN	$-1^{\circ}\text{C} \leq \Delta t \leq 1^{\circ}\text{C}$
HEATING	$\Delta t < -1^{\circ}\text{C}$

4.1.2 Cooling mode

4.1.2.1. Temperature control :16-31°C(61-88°F), and the operation of fan speed and vane position according to preset.

4.1.2.2. Compressor and process control

1). When $RT - ST \geq 1^{\circ}\text{C}$ the compressor starts up working, AC operates as customer preset;

2). When:

a. $RT - ST \geq 1^{\circ}\text{C}$, the compressor will start up operation once it stops 3 minutes, and unit without any other protection or failure code.

b. $RT - ST \leq -1^{\circ}\text{C}$, the compressor will stop working while it works for 6 minutes or more continuously.

c. $-1^{\circ}\text{C} < RT - ST < 1^{\circ}\text{C}$, the compressor keeps original state.

3). The compressor will also stop working while unit:

a. switched off.

b. under protection.

c. changed to fan mode.

4). Under normal operation, the compressor can be stopped by program until it works for 6 min at least once starts up.

5). In the process of unit operation, once the compressor ceased, it should be 3 min delay for the next procedure.

4.1.2.3. ODU Fan motor control:

1). While unit:

a. switched off.

b. under protection.

c. to the set temperature.

The compressor stops working, and the fan motor will stop operation after 50s delay.

2). When switch on unit on cooling mode, ODU fan motor will delay 5s after compressor starts up.

4.1.2.4. When ODU failure or stops for protection, IDU works as preset.

4.1.2.5. Anti-frosting protection

While condition a), b) matched, the AC unit transferred to anti-frosting protection.

a). After the compressor continual works for 6 minutes, and tested $\text{IPT} \leq 0^{\circ}\text{C}$ for 3 minutes continuously.

b). After the compressor continual works 6 minutes, and tested $\text{IPT} \leq -2^{\circ}\text{C}$ for 3 seconds continuously.

c). AC unit quit protection while $\text{IPT} \geq 10^{\circ}\text{C}$.

4.1.3 Dry mode

4.1.3.1. Temperature control: $16 \sim 31^{\circ}\text{C}$.

Fan speed: low

Vane blade position: as customer preset

4.1.3.2. When ODU failure or stops for protection, IDU works as preset.

4.1.3.3. Compressor and process control

1). When $\text{RT} - \text{ST} \geq 1^{\circ}\text{C}$, the compressor starts up working, AC operates as customer preset;

2). When

a. $\text{RT} - \text{ST} > 2^{\circ}\text{C}$, the compressor will start up operation once it stops 3 minutes, and unit without any other protection or failure code.

b. $\text{RT} - \text{ST} < -2^{\circ}\text{C}$, the compressor will stop working while it works 6 minutes or more continuously.

c. $-2^{\circ}\text{C} \leq \text{RT} - \text{ST} \leq 2^{\circ}\text{C}$, the compressor will stop working while it works for 6 minutes, after 4 minutes it will start working again.

3). The compressor will also stop working while unit:

a. switched off.

b. under protection.

c. changed to fan mode.

4). Under normal operation, the compressor can be stopped by program until it works for 6 min at least once starts up.

5). In the process of unit operation, once the compressor ceased, it should be 3 min delay for the next procedure.

4.1.3.4. ECO and sleep mode: Invalid.

4.1.3.5. Anti-frosting protection

While any of condition a), b), or c) matched, the AC unit transferred to anti-frosting protection.

a). After the compressor continual works for 6 minutes, and tested $\text{IPT} \leq 0^{\circ}\text{C}$ for 3 minutes continuously.

b). After the compressor continual works 6 minutes, and tested $\text{IPT} \leq -2^{\circ}\text{C}$ for 3 seconds continuously.

c). If the unit works as 6 minutes ON, 4 minutes OFF, and tested $\text{IPT} \leq 0^{\circ}\text{C}$ for 3 minutes continuously.

d). AC unit quit protection while $\text{IPT} \geq 10^{\circ}\text{C}$.

4.1.4 Heating mode

4.1.4.1. Temperature control: $16 \sim 31^{\circ}\text{C}$.

4.1.4.2. Compressor and process control.

1). When $\text{ST} - \text{CRT} \geq 1^{\circ}\text{C}$, the compressor starts up working, AC operates as customer preset;

2). When

a. $\text{ST} - \text{CRT} \geq 1^{\circ}\text{C}$, the compressor will start up operation once it stops 3 minutes, and unit without any other protection or failure code.

b. $\text{ST} - \text{CRT} \leq -1^{\circ}\text{C}$, the compressor will stop working while it works 6 minutes or more continuously.

c. $-1^{\circ}\text{C} < \text{ST} - \text{CRT} < 1^{\circ}\text{C}$, the compressor keeps original state.

3). The compressor will also stop working while unit:

a. switched off.

b. under protection.

c. changed to fan mode.

- 4). Under normal operation, the compressor can be stopped by program until it works for 6 min at least once starts up.
- 5). In the process of unit operation, once the compressor ceased, it should be 3 min delay for the next procedure.

4.1.4.3. IDU time delay: When compressor stops or unit switches off while in heating mode, IDU fan motor keeps working for 50 seconds.

4.1.4.4. ODU Fan motor control:

1). While unit:

- a. To be switched off
- b. Under protection.
- c. To the set temperature

The compressor stops working, and the fan motor will stop operation after 50s delay.

- 2). When switch on unit on heating mode, ODU fan motor will start up 3s earlier before compressor starts up.
- 3). In the process of defrosting, the fan motor will stop operation 39s delay after compressor stopped.
- 4). defrosting finish, the compressor stops operation, the fan motor will start working simultaneously.

4.1.4.5. 4-way valve control

- 1). On Cooling/Dry/Fan mode, 4-way valve: **OFF**,
when unit switched on heating mode, 4-way valve: **ON**.
- 2). When heating mode switched off, or changed from heating to other modes, the 4-way valve will be **OFF** 2 min delay after the compressor stops working.
- 3). The 4-way valve will be **OFF** 2 min delay while unit stops working caused by any kinds of protection.
- 4). In the process of defrosting, 4-way valve will be **OFF** 39s delay after compressor stopped.
- 5). When defrosting finish, the compressor stops operation, 4-way valve will be **ON** after 20s delay.

4.1.4.6. Defrost

Note:

t1: The compressor continuous works time.

t2: The compressor accumulated working time.

t3: The tested max value of "IPT-RT" after compressor works for 8 minutes till to current.

ΔT : The current value of "IPT-RT" while compressor works. (*the instantaneous value*)

1). When AC unit works matching a), b) and c):

- a. $t1 \geq 120$ minutes,
- b. $t2 \geq 45$ minutes,
- c. $t3 \geq \Delta T + 4^{\circ}\text{C}$.

it triggers to defrosting.

Defrosting start up: Compressor stops working, and re-starts operation after 44s delay;

2). AC unit quit defrosting automatically after 10 minutes operation.

Defrosting end off: Compressor stops working, and re-starts after 25s delay;

4.1.4.7. Cold air prevention:

This function intends to prevent cold air from being discharged when the heating operation starts up

1). IDU fan motor cold air prevention

① .When $RT < 24^{\circ}\text{C}$:

- a. If $ITP > 31^{\circ}\text{C}$ while compressor works for 5 min, fan motor will work according to preset fan speed.
- b. If $IP T \leq 31^{\circ}\text{C}$, fan motor stops working while compressor operation within 2min, if $IP T \geq 27^{\circ}\text{C}$, fan motor works in low speed 2 min, then change to preset speed.

② When $RT \geq 24^{\circ}\text{C}$:

- a. Within 2 min after compressor starts up working, once $IP T > 27^{\circ}\text{C}$, fan motor will change to preset speed.
- b. After compressor starts working 2 min, fan motor changes to preset speed directly.

2). Vane blade operation for cold air prevention.

a. If the vane works on status of **non-swing**: While IPT increase to a special degree, the vane blade will change the position to:

- ① Default angle; or
- ② Previous position; or
- ③ Customer preset.

The above operation can work one time only, when compressor stops working, the unit will inspect the function once again.

b. If the vane works on **swing**: When IDU quits from cold air prevention, vane operate as preset.

4.1.4.8. "8°C" Heating

When function effective, it will make sure the room temperature not lower than 8°C, unit works on heating mode.

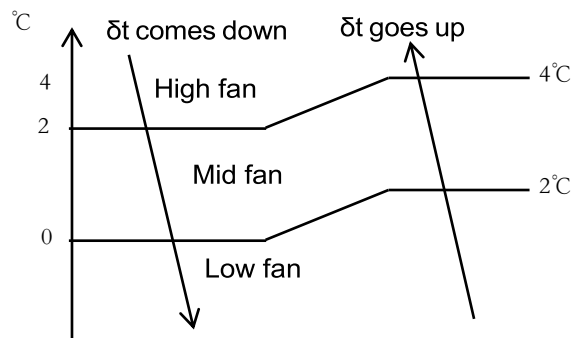
4.1.5 Fan mode

4.1.5.1. The temperature setting: 16~31°C(61-88°F),

Fan speed and vane position: as preset. The function for remote controller 58E only.

4.1.5.2. For above function, when unit preset to auto fan mode, the fan motor will change its operation speed based on the temperature difference of ambient and preset temperature.

4.1.5.3. ODU always OFF.



4.1.6 Timer

The unit has times control, when the unit meet the timer preset, it will be switched on or off automatically.

4.1.6.1 TIMER ON

- 1). TIMER ON can be set only when the air conditioner is OFF.
- 2). Press TIMER on the remote control ONCE to enter time setting.
- 3). Press "▲" or "▼" to set the time for unit to start working.
- 4). Set other function as MODE, FAN SPEED, SWING etc.
- 5). Press TIMER ONCE AGAIN to confirm the TIME ON setting

4.1.6.2 TIMER OFF

- 1). TIMER OFF can be set only when the air conditioner is ON.
- 2). Press TIMER on the remote control ONCE to enter time setting.
- 3). Press "▲" or "▼" to set the time for unit to stop working.
- 4). Press TIMER ONCE AGAIN to confirm the TIME OFF setting

Note: In case of no selection for the time setting within 10s, the timer function will be OFF automatically.

4.1.7 Sleeping mode

While AC works in sleeping mode, the light of POWER SUPPLY and SLEEP always ON, and the temperature display will be OFF after 15s.

In this mode, the AC unit works according to the SLEEP CURVE as designed.

Sleeping mode the unit can work 10 hours continuously, after that it will quit from this mode and work as previous presetting.

4.1.8 Emergency switch

When the EMERGENCY switch is pressed one time, COOLING mode is selected and if it pressed again within 3s, HEATING mode selected, while press once again, the unit will be switched off.

When the remote controller out of function, batteries lost power, for example, the EMERGENCY button in the front of indoor unit can be used for function test.

NOTE: Do not press the EMERGENCY switch during normal operation.

4.1.9 Auto-restart function

While air conditioner is operating in one mode, all of its operation data, such as working mode, preset temperature etc. would be memorized into IC by main PCB. If power supply cut off due to reasons and recover again, the AUTO-RESTART function will set synchronously and the air conditioner would work at the same mode as before.

Note: The function setting:

Within 3 min while power on unit, set the unit on cooling mode, set temperature 30°C, and mid fan speed, press the ECO button 10 times within 8s, the AUTO-RESTART will be activated.

4.1.10 Over heat protection for Heating mode

- 1). If:
 - a. $IPT \geq 62^{\circ}\text{C}$, compressor and ODU Fan stop working for over heat protection.
 - b. $IPT \geq 54^{\circ}\text{C}$, ODU Fan stops working. IDU fan motor and compressor keeps original state.
- 2). When $IPT \leq 48^{\circ}\text{C}$, the compressor and ODU Fan restart works to quit overheat protection.
- 3). If unit have 6 times Overheat Stop-working Protection continuously, or tested $IPT \geq 62^{\circ}\text{C}$ for 3mins continuously while compressor stop working, this protection can't be recovered unless press ON/OFF button, and unit will show failure code "P7".

Note:

In the process of operation, once the compressor runs continuously more than 6 min, the counter of overheat stop-working protection will be reset to zero and start a new counting process.

The failure and times for protection will eliminate immediately once the unit to be switched off, on fan mode or changed to be heating mode from others.

4.1.11 Complementary

4.1.11.1 Energy saving (ECO)

Function effective on Cooling and Heating mode only.

On cooling mode, the set temperature range from $26^{\circ}\text{C}(79^{\circ}\text{F})$ to $31^{\circ}\text{C}(88^{\circ}\text{F})$.

on heating mode, from $16^{\circ}\text{C}(61^{\circ}\text{F})$ to $25^{\circ}\text{C}(77^{\circ}\text{F})$.

4.1.11.2 TURBO

Function effective on Cooling, Heating, Fan and Auto modes, and fan speed operates on highest.

4.1.12 Calibration Test Mode:

Within 3 min while indoor unit switch on, and set the unit as:

- 1) Cooling mode.
 - 2) set temperature to 30°C .
 - 3) Mid-fan speed.
- by press ECO button 7 times within 8s, the unit will change to calibration test mode, and the buzzer sounds 3 times.